

- 8 Fig. 8.1 shows a simplified circuit diagram of a laptop adaptor connected to the chargeable battery of a laptop. The adaptor consists of a step-down transformer with a single diode.

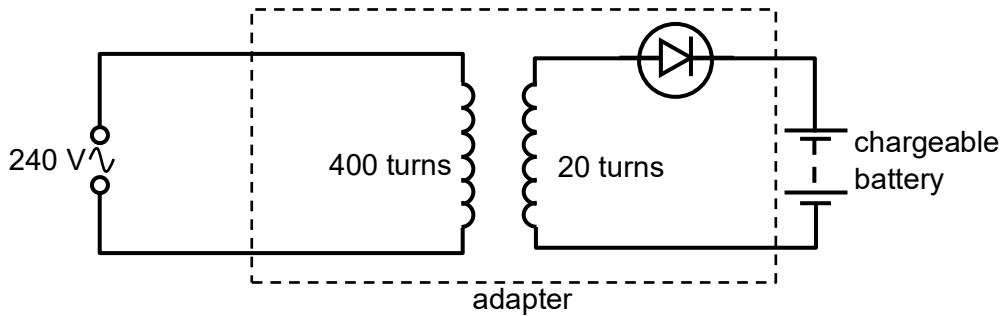


Fig. 8.1

- (a) Calculate the peak voltage across the secondary coil and state one assumption made in performing the calculation.

$$\frac{N_s}{N_p} = \frac{V_s}{V_p}$$

$$V_s = 20/400 (240) = 12 \text{ V}$$

$$\text{Hence } V_o = \sqrt{2} \times 12 = 17 \text{ V}$$

peak voltage = V [1]

Assumption: ... the transformer is ideal/100% efficiency. [1]

- (b) Explain why, during the charging process, the power delivered to the secondary coil is twice the power delivered to the chargeable battery.

The voltage step down to the secondary coil is sinusoidal but with the diode in place,
 the voltage across the battery is half wave rectified . This imply that the time that the [1]
 battery get charged is only half of the time voltage is step down across the secondary
 coil. Hence the energy received by the battery is reduced by half.

- (c) Hence show that the voltage rating of the chargeable battery is 8.5 V.

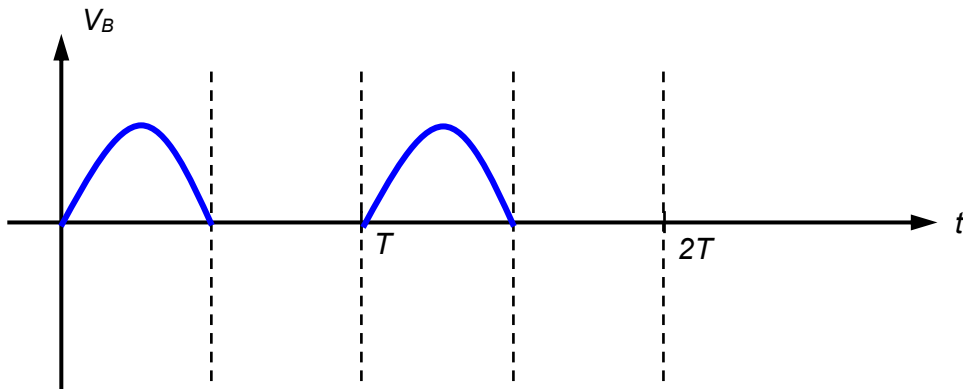
Without diode , assume power is P and $P = (12)^2/R$

With diode , power delivered = $P/2 = V_{\text{rms}}^2/R$

Hence $V_{\text{rms}} \sqrt{(144/2)} = 8.5 \text{ V}$

[1]

- (d) Sketch on Fig. 8.2 the variation of potential difference V_B across the chargeable battery with time t . It is given that T is the period of the alternating current.



[1]

Fig. 8.2

- (d) Hence sketch on Fig. 8.3 the variation of the potential difference across the diode V_D with time t . Label appropriate values. It is given that T is the period of the alternating current.

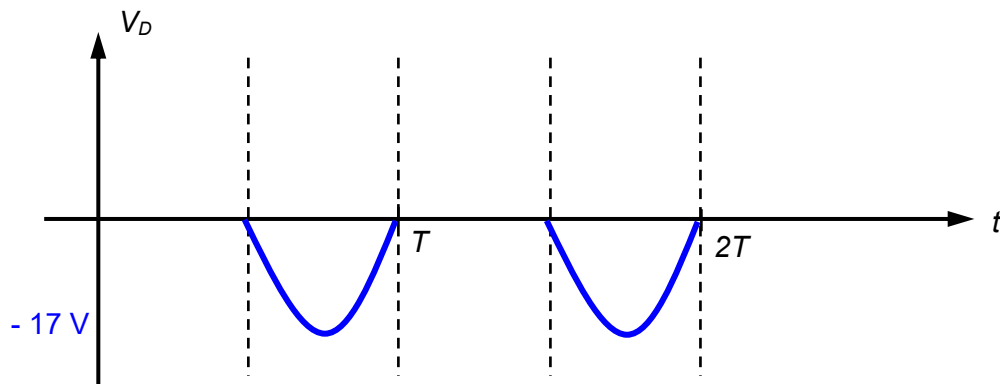


Fig. 8.3

[2]

[Total: 7]